Effects of Applying Learning Theory to Brief Experimental Analyses of Reading Difficulties

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Abstract

In a Response to Intervention model, students who are struggling to succeed with quality core instruction and supplemental interventions are provided an individualized intervention. The current study examined using a brief experimental analysis (BEA) to identify a specific instructional strategy to accelerate reading growth for individual students. The model was implemented with 17 students from 5 schools in a large urban district in the Midwestern United States. The effectiveness of interventions to address acquisition, retention, and generalization were tested with a multi-element design within the BEA and the intervention that led to the largest effect was implemented four times per week for a period of several weeks. Slopes of growth were computed the monitoring data with ordinary least squares and compared pre-BEA and post-BEA. A Wilcoxon signed-rank test found a reliable change in slopes from pre-BEA to post-BEA for students in grades 1-3 (z = -2.67, p < .01) and for students in kindergarten (z = 2.02, p < .05).

Introduction

- There is a growing array of evidence-based interventions for struggling readers, but no single approach is effective for all students (Fuchs et al., 2001; Fuchs & Fuchs, 2005).
- Brief experimental analysis (BEA) has proven to be a useful tool in the problem solving process for students for whom generally effective interventions were not effective. BEAs are short empirical approaches to test hypotheses regarding the function of performance deficits.
- Daly, Witt, Martens and Dool (1997) suggested using BEAs with academic performance deficits to test (a) lack of motivation, (b) insufficient practice, (c) inadequate support and feedback, (d) novel instructional materials, and (e) difficult instructional materials as hypotheses for student failure.
- BEAs have been shown to be effective in identifying appropriate reading interventions for kindergarteners (Petursdottir et al., 2009), for students in the primary grades (Daly et al., 1999; 2006), and for letter formation interventions for 2nd graders (Burns, Ganuza, & London, 2009).
- Learning theory suggests that students must first initially acquire a skill, then practice it to proficiency, then recall it at a later time, and then finally generalize the skill to different times and settings (Alberto & Troutman, 2006; Binder, 1996; Haring & Eaton, 1978).
- Previous research examined the Daly et al. (1997) hypotheses for student failure, but did test hypotheses based on learning theory.
- The purpose of the current study was to examine the effectiveness of developing intensive interventions with BEAs based on learning theory for 2nd and 3rd grade students who did not respond well to small-group interventions. It was implemented as part of the Path to Reading Excellence in School Sites (PRESS) project.

Method

PRESS
Path to Reading Excellence in School Sites

Tier II Problem Solving

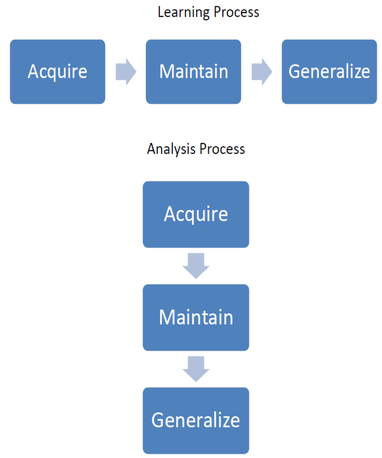
Student Name:_____Grade:___Teacher:_____
Intervention Group:____School_____Literacy Assistants:_____
Current Intervention:_____Past Interventions: _____
Data Meeting where identified (date):_____

Recent Scores:

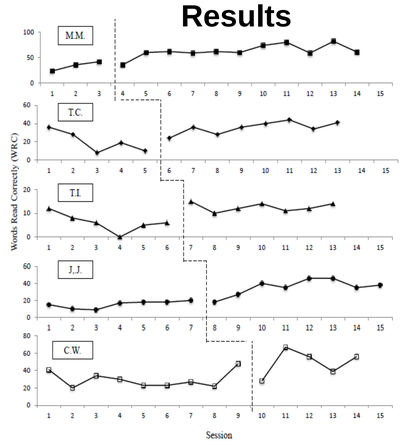
	Skill__	Skill__	Skill__	Skill__	GOM__	GOM__	GOM__	Growth Slope	Target Benchmark for GOM
Type									
Score									

1. Is the most recent GOM score below the upcoming seasonal target but the GOM slope (growth) is at or above the criterion?
   a. If Yes, continue supplemental support/intervention
   b. If No (GOM score and slope are both below criterion), examine skill measure data
2. Are skills being acquired using current strategy?
   a. If Yes, continue with current strategy
   b. If No, adjust intervention within level of support to match skill need
3. Is mean instructional fidelity during intervention with the student below 90%?
   a. If YES, improve instructional fidelity.
4. Is student showing motivation difficulties (e.g., resisting participation)?
   a. If YES, provide incentives
5. Is student attendance below 95% during last instructional period (1 month)?
   a. If Yes, address attendance with building administration
6. Are behavioral difficulties leading to student missing intervention more than 3 times during the last instructional period (1 month)?
   a. If Yes, address behavior difficulties with building administration
7. Has the student received intervention with good fidelity for less than 9 weeks?
   a. If YES, implement intervention for up to 9 weeks
8. Has the student received intervention for less than 20 min four times per week for the past instructional period (1 month)?
   a. If YES, increase time and sessions
9. If NO to Questions 3-8 above, move to Brief Experimental Analysis (BEA)/Tier III Intervention

PLAN: (Date)



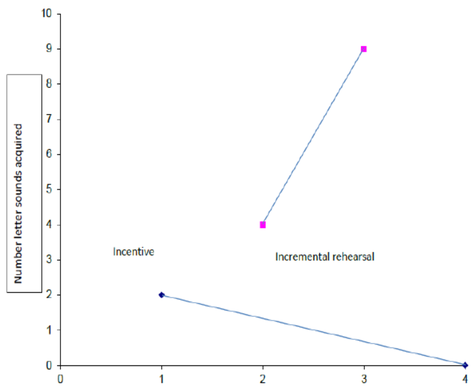
Results



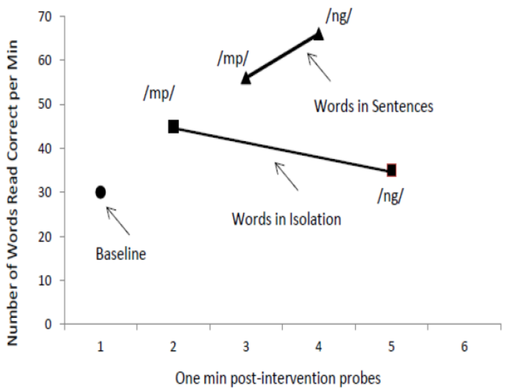
Students	Measure	# of Weeks Pre BEA	Pre BEA Slope	# of Weeks Post-BEA	Post BEA Slope	Change in Slope
S. I.	WRC	20	0.25	2	8.00	7.75
T. I.	WRC	12	-0.64	8	0.55	1.19
K. M.	WRC	10	1.50	14	1.68	0.18
L. F.	LSC	22	-0.15	8	0.12	0.26
M. M.	WRC	6	3.00	8	3.43	0.43
T. C.	WRC	10	-3.05	9	3.03	6.08
U. R.	WRC	16	0.07	7	0.46	0.39
J. J.	WRC	14	0.71	9	2.78	2.07
K. W.	WRC	8	0.90	8	1.06	0.16
K. Z.	LSC	20	1.32	2	8.00	6.68
S. Y.	WRC	8	-0.25	12	0.08	0.33
C. W.	WRC	18	0.11	6	1.77	1.66
S. S.	WRC	18	0.44	6	3.03	2.59
X. M.	WRC	6	0.00	6	-0.40	-0.40
A. S.	LSC	22	0.29	9	1.08	0.80
F. M.	LSC	14	0.82	7	2.93	2.11
S. D.	LSC	12	0.23	8	2.52	2.30

		Pre BEA	Post BEA	
	n	Mean Slope	Mean Slope	Signed Ranks
Letter-Sound Fluency	05	.77	2.93	Z = 2.02*
Oral Reading Fluency	12	.25	1.92	Z = 2.67*

Acquisition example



Generalization example



Discussion

- **Student rate of growth reliably increased after changing the intervention from the BEA.**
- **The acquisition-maintain-generalization sequence provide a potentially useful framework to conduct BEAs.**
- **Future research could examine the relative utility for BEA of the sequence studied here and the Daly et al. (1997) hypotheses for student failure.**
- **An experimental design was used (multi-element single-case design) to identify the appropriate intervention within the BEA, but the comparison of intervention effectiveness was not experimental and suggests that additional research is needed and warranted.**

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TARGET

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